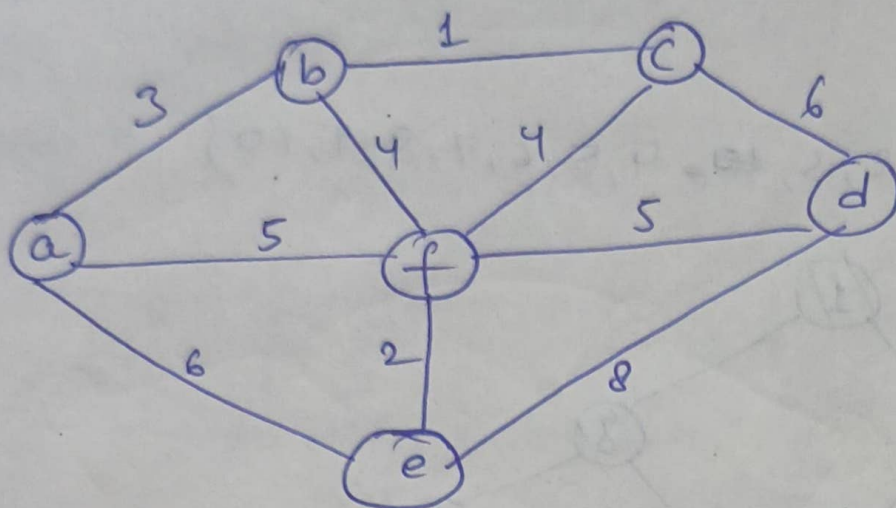


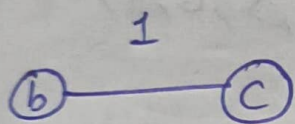
Kruskal Algorithm:-

1.



$|V| = 6$, $|E| = 10$, 5 edges
 bc ef ab bf cf af de ae cd de
 1 2 3 4 4 5 5 6 6 8
 ✓ ✓ ✓ ✓ ✗ ✗ ✓ ✗ ✗ ✗

i,



no cycle ✓

(a)

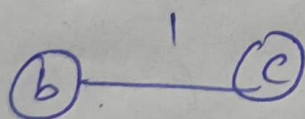
(f)

(d)

(bc)

(e)

ii,



no cycle ✓

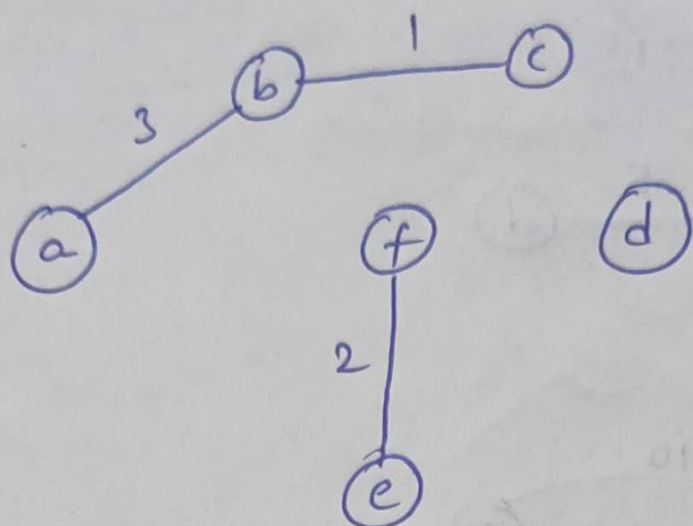
(ef)

(a)

(d)

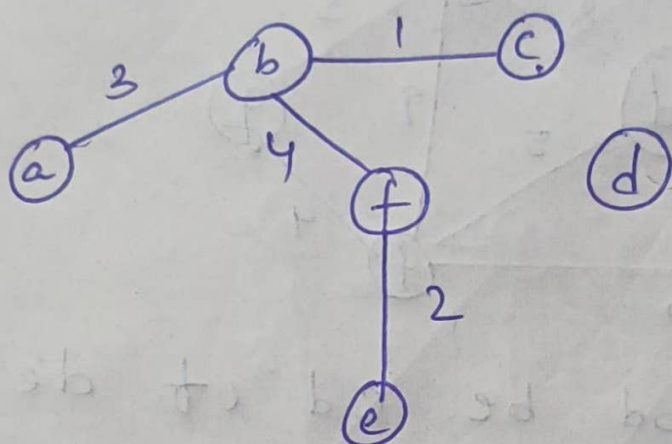
(f)
(e)

(iii)



no cycle ✓
(ab)

(iv)

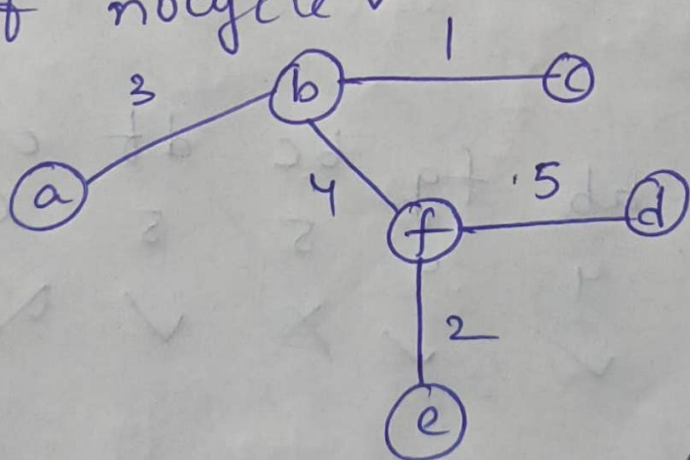


no cycle ✓
(bf)

(v) cf forms a cycle so skip

(vi) af forms a cycle so skip

(vii) df no cycle ✓



(viii) ae forms a cycle so skip.

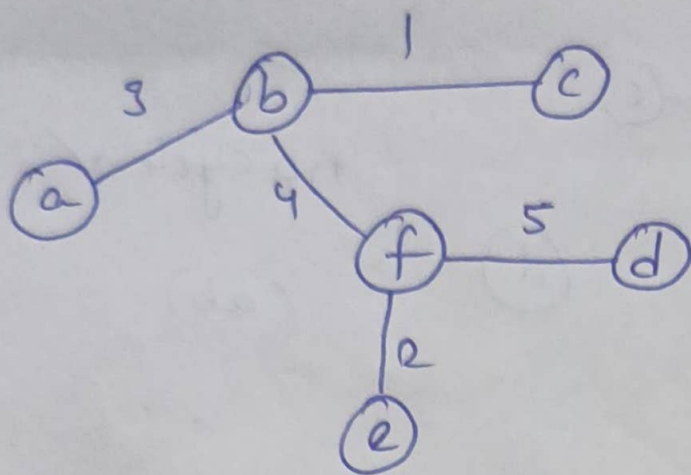
(ix) cd forms a cycle so skip.

(x) de forms a cycle so skip

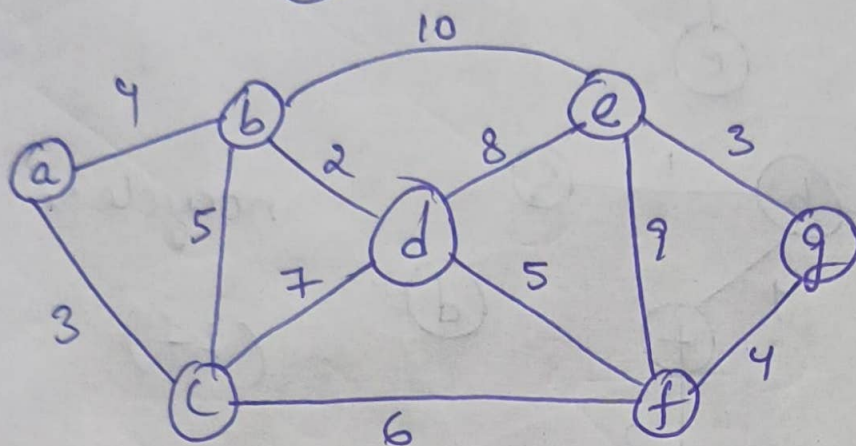
minimum spanning tree weight

$$= 1 + 2 + 3 + 4 + 5$$

$$= 15$$



2.



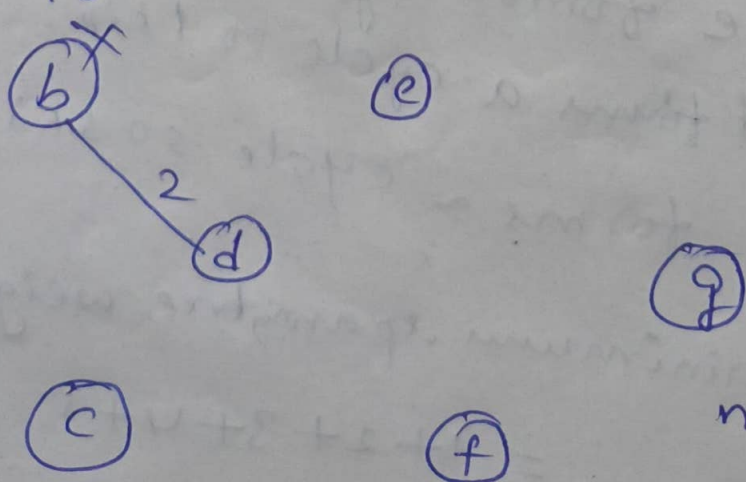
ab	ac	bc	bd	be	cd	cf	de	df
4	3	5	2	10	7	6	8	5

ef	eg	fg
9	3	4

sorted :-

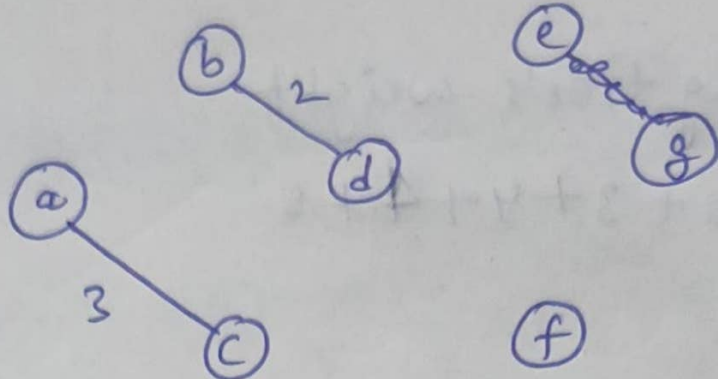
bd	ac	eg	ab	fg	bc	df	cf	cd
2	3	3	4	4	5	5	6	7
✓	✓	✓	✓	✓	✗	✓	✗	✗

de	ef	be
8	9	10
✗	✗	✗



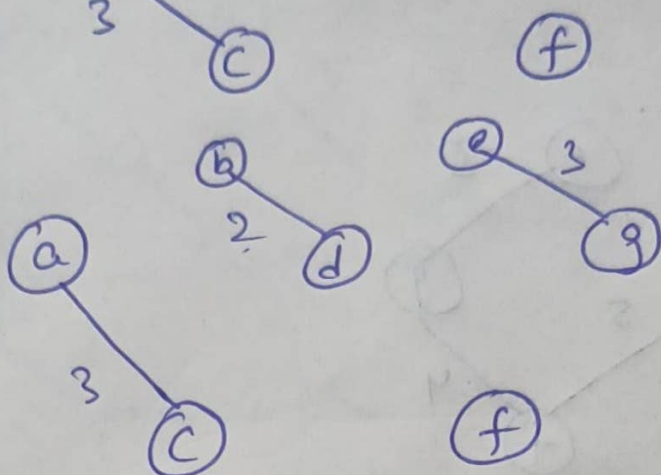
nocycle ✓

(ii)



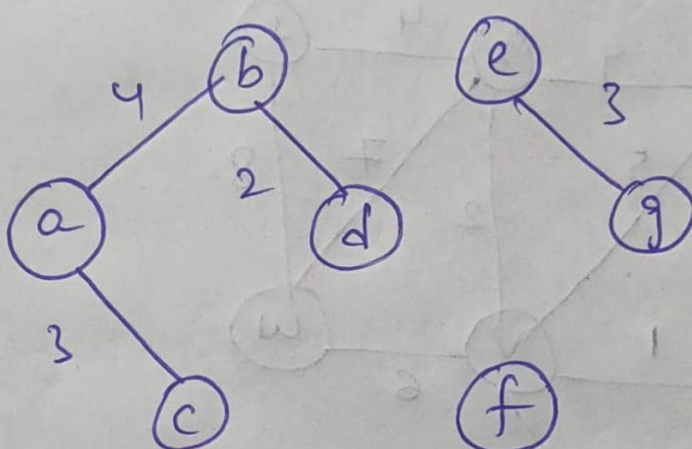
no cycle ✓

(iii)



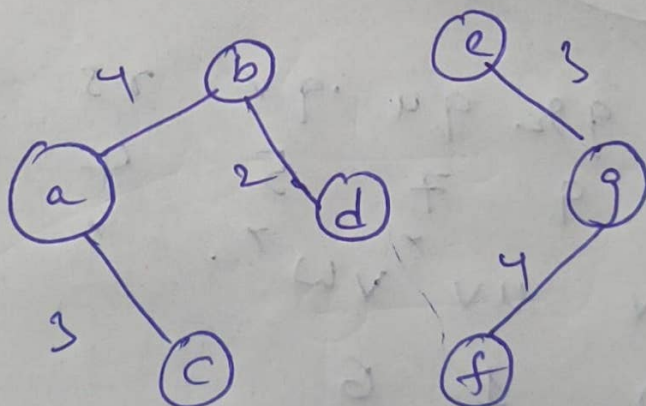
no cycle ✓

(iv)



no cycle ✓

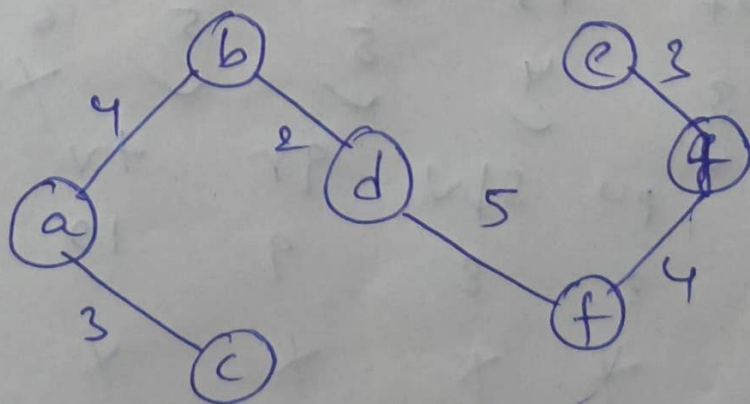
(v)



no cycle ✓

(vi) bc forms a cycle so skip

(vii)



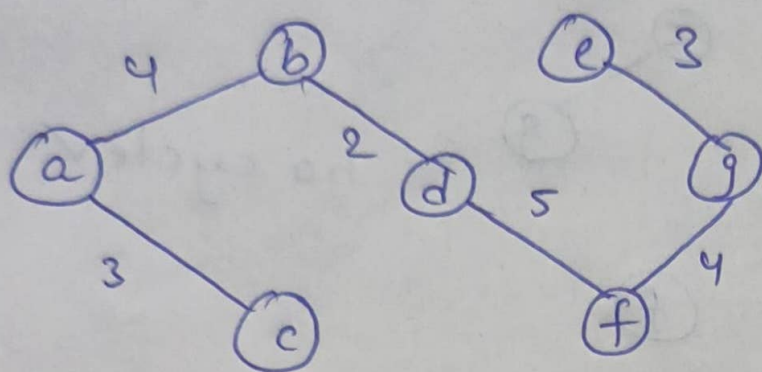
no cycle

(viii) cf, cd, de, ef, be form a cycle so skip

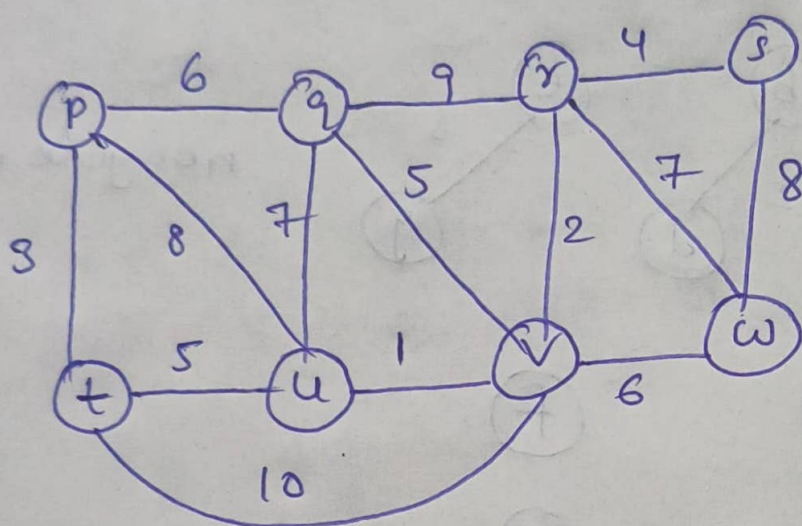
Minimum spanning tree's weight

$$= 2 + 3 + 3 + 4 + 4 + 5$$

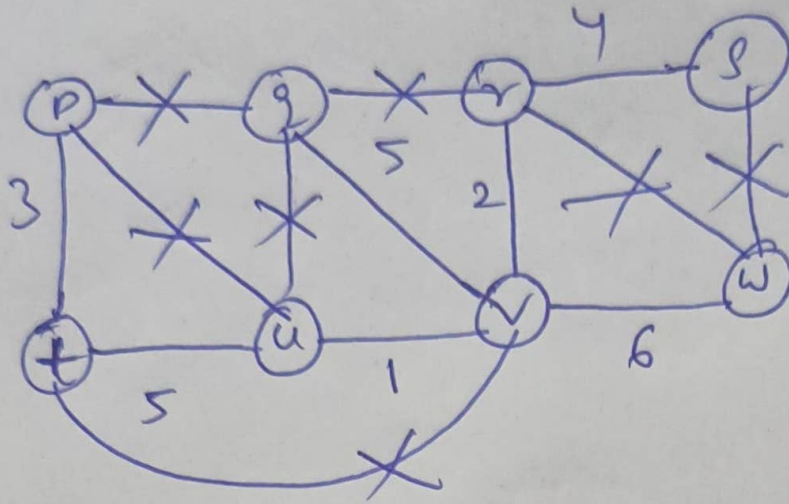
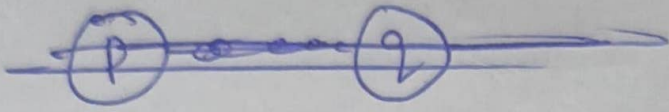
$$= 21$$



3)



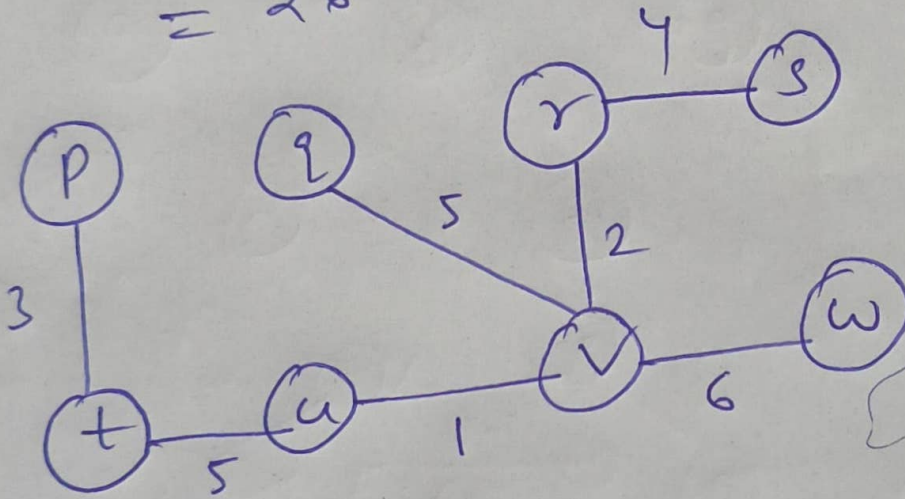
pt 3, pu 8, pq 6, qr 9, qu 7, qv 5, rs 4, rw 7, rv 2
 tu 5, tv 10, uv 1, vw 6, sw 8, x 7, x 7
 sorted: -
 uv 1 ✓, rv 2 ✓, pt 3 ✓, rs 4 ✓, qv 5 ✓, tu 5 ✓, pq 6 ✗, vw 6 ✓
 qu 7, rw 7, pu 8, sw 8, qr 9, tv 10
 7, 7, 7, 7, 7, 7



Minimum spanning tree's weight

$$= 1 + 2 + 3 + 4 + 5 + 5 + 6$$

$$= 26$$



MST